Vijaya Sreyas Badde

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.85
Narayana Junior College	Class XII (TGBIE)	2024	86.00%
Narayana High School	Class X (TS-SSC)	2022	8.70%

SCHOLASTIC ACHIEVEMENTS

Inter-IIT Tech Meet 14.0

- Selected to represent IIT Hyderabad at the Inter-IIT Tech Meet 14.0 as part of a 4-member team tackling the GDAI Problem Statement.

POSITIONS OF RESPONSIBILITY

Game Development Core: Glitch (GameDev and eSports Club of IIT Hyderabad)

Overall Coordinator: Glitch (GameDev and eSports Club of IIT Hyderabad)

WORK EXPERIENCE

VR Development Intern | *Department of Design, IIT Hyderabad*

- Built CAD-import and improved mechanism-joint interaction tools for a VR sketching application on Meta Quest.
- Contributed to a separate research project involving harmonic tidal analysis and automated reporting.

SKILLS

Languages - Python, C, SQL, C# (Unity)

Frameworks & Libraries - NumPy, Pandas, Matplotlib, OpenCV, Scikit-learn, PyTorch, TensorFlow

Tools & Technologies: Git, GitHub, Linux, Jupyter Notebook, LaTeX, Unity

PROJECTS

3D Sketching Interface for Mechanism Behaviour Analysis using AI (SIMBA 2.1) | *Department of Design, IITH*

- Built a runtime STL import pipeline that parses mesh data off the main thread and instantiates a selectable, collider-backed 3D model.
- Built a runtime transform gizmo (move/rotate/scale) for manipulating imported models without pre-built mesh assets.
- Implemented non-destructive semantic grouping/ungrouping of hand-drawn strokes without altering underlying stroke geometry.
- Built a reusable stroke geometry/analysis layer and a frame-spread processing queue to keep background tasks off the main thread.
- Extended mechanism-design tools (spring, fixed, prismatic, hinge joints) with anchor lifetime tracking to prevent stale-reference crashes.

Rock-Paper-Scissors Image Classifier | *Programming for AI* | (Link)

- Built a custom PyTorch Dataset/DataLoader pipeline for RPS classification, with transforms and a reproducible 80/20 train-test split.
- Implemented depthwise convolutions with hand-crafted kernels (blur, sharpen, edge detection) to visualize feature

extraction.
- Designed and trained a CNN with convolutional, max-pooling, and global average pooling layers, tracking accuracy across epochs.

Rowmance | *Glitch, the GameDev and eSports Club Club of IIT Hyderabad* | (Link)

- Designed and implemented the core multiplayer architecture for a cooperative Unity game using Unity Netcode for GameObjects.
- Developed networked gameplay systems including session management, player/pause synchronization, and disconnect handling.
- Built checkpoint synchronization, persistent save/load, and cross-platform input support for multiplayer sessions.

Fair Bayesian Active Learning for Credit Risk | *AI for Humanity* | (Link)

- Developed a Fair Active Learning system on a Bayesian logistic backbone to correct sampling bias in a live credit-risk labelling loop.
- Designed and evaluated 4 label-selection strategies, raising minority representation from ~9% to ~50% without degrading accuracy or F1.
- Implemented custom Balanced Gradient Descent and Monte Carlo fairness-gain scoring to optimize quality and demographic parity.
- Produced 18 diagnostics covering representation, calibration (ECE, Brier score), and per-group TPR/FPR gap across all strategies.

Wordle Strategy Simulation & Optimization | *Programming for AI* | (Link)

- Implemented Wordle's feedback logic (green/yellow/gray tiles) and analyzed letter-frequency distributions across a 3,103-word dataset.
- Estimated guess-outcome probabilities via Monte Carlo simulation and a greedy candidate-filtering strategy to simulate full games.
- Formulated and solved a convex optimization problem with cvxpy for the optimal trade-off between green- and yellow-tile probabilities.

Dual-World Puzzle Platformer | *Inter-IIT Tech Meet 14.0* | (Link)

- Developed a Unity-based game combining 3D exploration and a 2D puzzle platformer with interconnected progression.
- Implemented gameplay mechanics including color-switching puzzles, checkpoints, and interactive environmental objects.
- Designed and built multiple game levels, balancing puzzle difficulty, pacing, and exploration flow across the 3D and 2D worlds.

RELEVANT COURSES

Artificial Intelligence: Introduction to Modern AI, Introduction to Classical AI, Programming for AI, Foundations of Machine Learning, Deep Learning, AI for Humanity

Computer Science: Introduction to Programming, Data Structures & Applications, Algorithms, Database Management Systems-I

Mathematics: Discrete Mathematics, Matrix Theory, Calculus I & II, Series of Functions, Vector Calculus, Differential Equations, Probability & Random Variables, Introduction to Metric Spaces, Introduction to Statistics, Optimization I & II, Numerical Analysis, Linear Systems & Signal Processing